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MASTER DATA MANAGEMENT SYSTEM

-- DATA ORGANIZATION MODULE --

László Kerekes D0PE82 – Report 2023. 04.21.

Supervisor: Dr. István Vassányi

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PROBLEM RAISING

- The more data sources, the more knowledge ÿ higher profit
- BUT! difficult connectivity ÿ lack of common keys, different structure, different quality

Example:

Marketing system

Customers registered on the web	
Email address	private.mail@email.com
Name	John Example
Employer name, address	Appel Corp, One Appl Prkwy
Web browsing	4 hours

Sales system

Telephone sales data	
Name	Mr. Joe Example
Phone number	555-12345-000 (mobile)
Employer	Apple
Amount sold	$150

"fuzzy" matching

"fuzzy" matching

Master data (checked reference)

Customers	
Name	Mr. John Example
Email address	my.company.mail@email.com
Employer name	Apple Inc.
Employer address	One Apple Park Way; Cupertino, CA 95014, USA
Phone number	555-56789-000 (company landline)
Classification	VIP

Expected statement

Customer statement	
Name	Mr. John Example
Employer	Apple Inc.
Classification	VIP
Uploaded on the web Correlation between hours and sales	0.65

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SPECIFICATION

My goal: to implement a master data management web application that supports the management and linking of data records coming from heterogeneous source systems

Functions:

- input data and master data management by topic
- linking data records • manual verification

  of proposed links
- display related data
- visualizing the goodness of associations

Existing solutions:

- MagellanER – traditional, rule-based ÿ a lot of manual work
- DeepER – convolutional and feedback networks ÿ CNN and RNN limitations
- Ditto – transformer-based network ÿ state-of-the-art

Uniqueness of my method: a

master data management framework that uses a transformer-based neural network for record association

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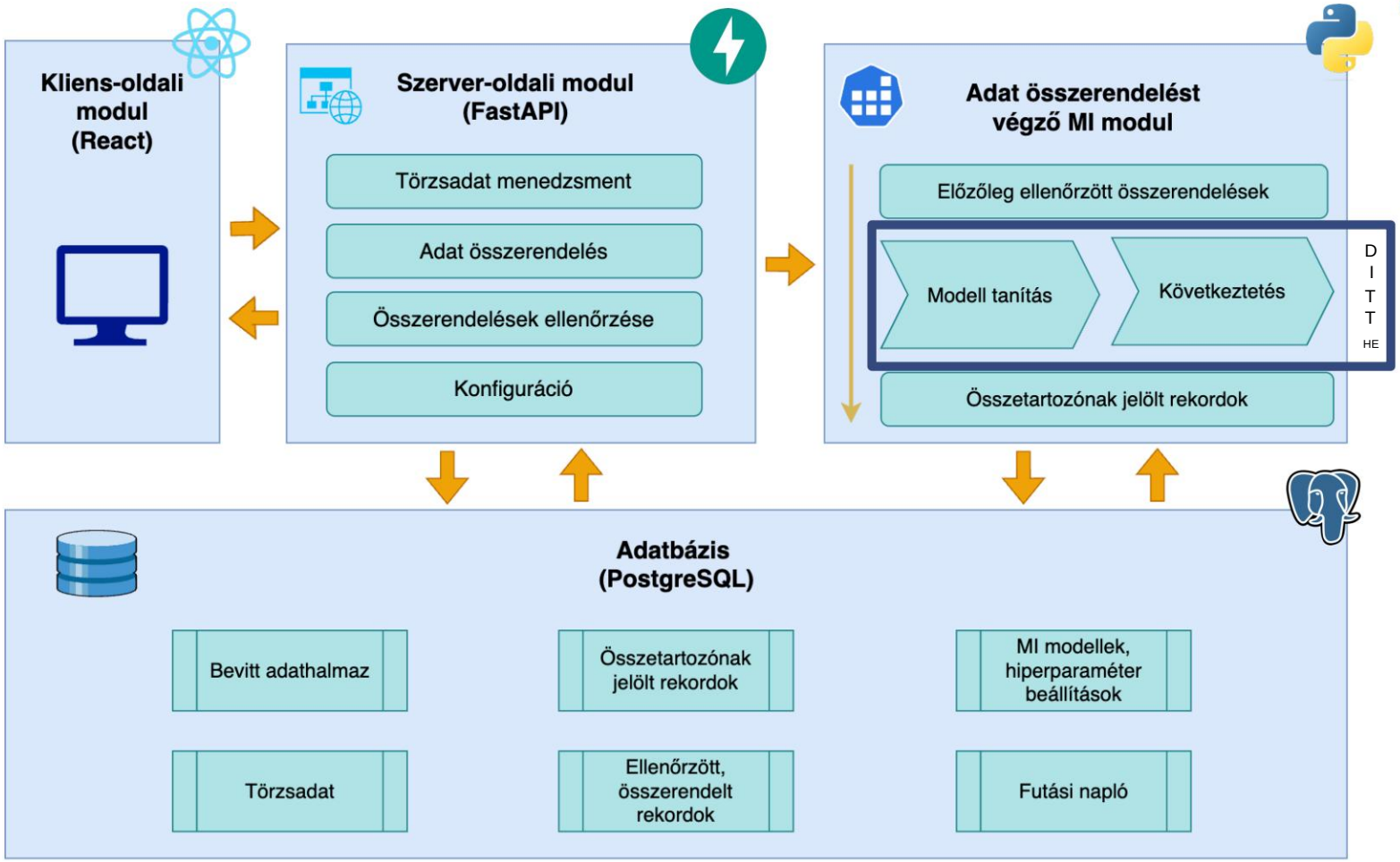
SYSTEM ARCHITECTURE



Data binding – Ditto

- Based on a study on entity matching* developed algorithm
- Transformer network-based solution based on a pre-trained language model: BERT
- Depending on the version: 6, 12, 24 layers, large neural network trained on text corpora
- Transformer advantages:
  - pre-trained, just fine-tune • handles records at once • processes text in two directions • can see text-wide connections, patterns

*Li & Li: Deep Entity Matching with Pre-Trained Language Models (2020)

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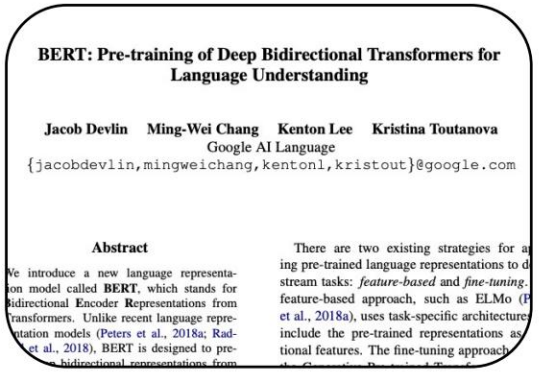
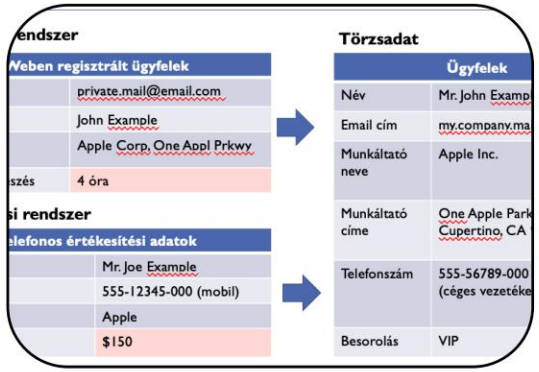
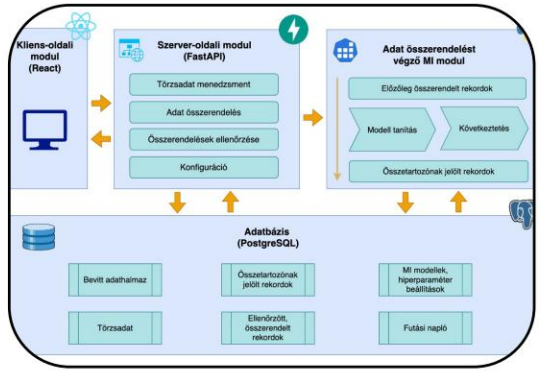
PLANNING



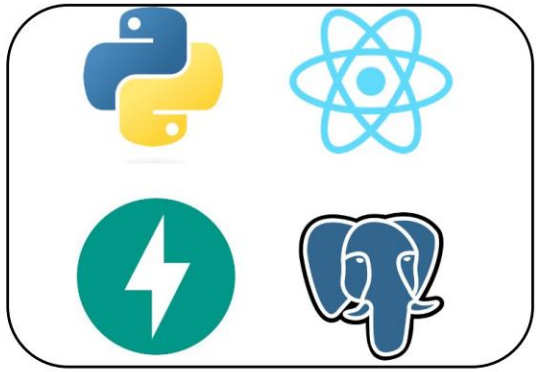
I became familiar with the **literature on** data management and data association.



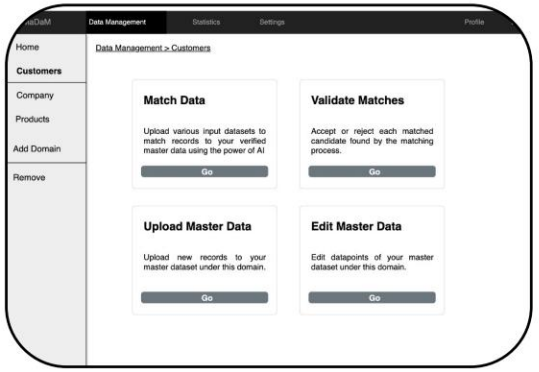
I defined the functional **requirements**



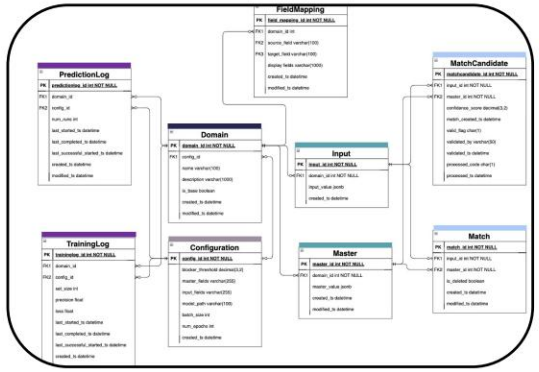
I designed the system **architecture.**



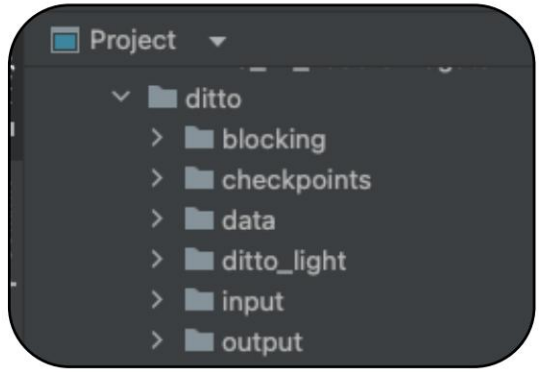
I selected the **technologies** that best suited my goals.



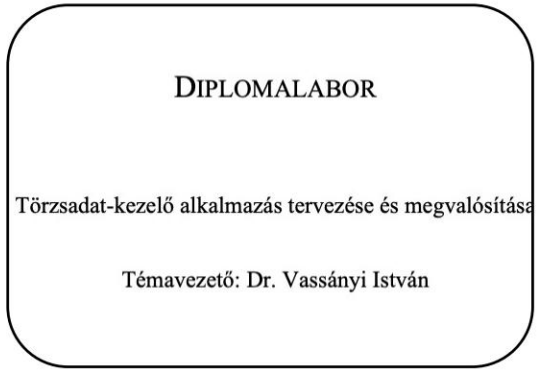
I designed the **graphic. interface** main screens and menu items



I created **the database .** its relational model, tables and their fields



I installed Ditto package on my own computer



I have prepared the **document** containing the system plan.

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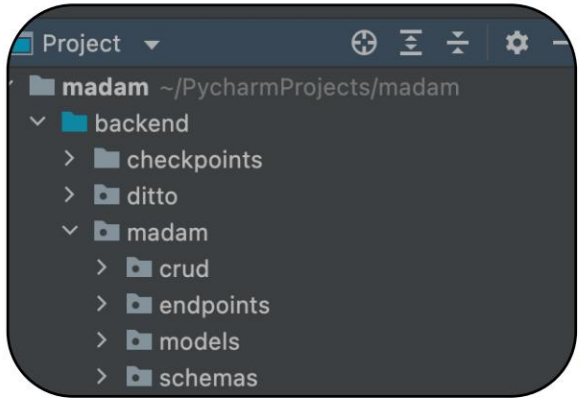
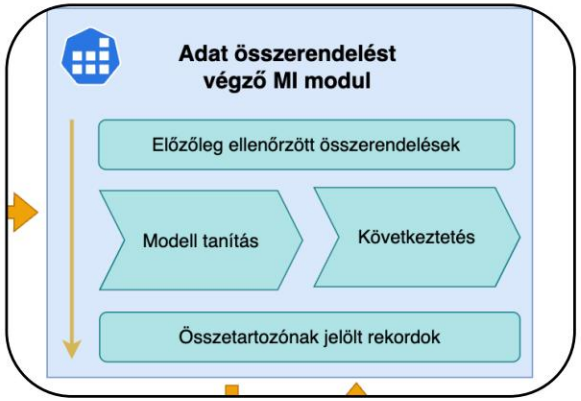
IMPLEMENTATION

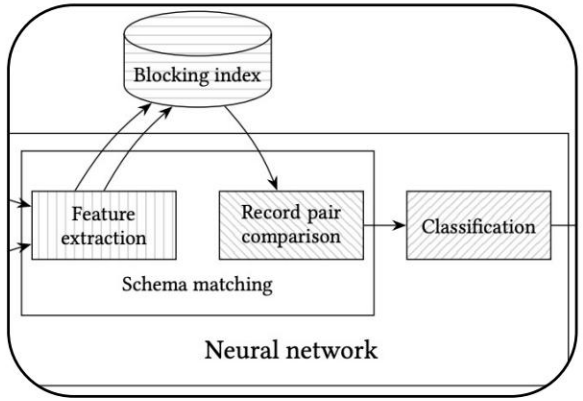


Framework

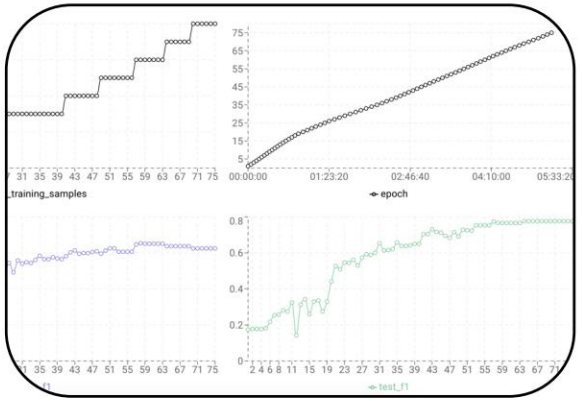
implementation



Embedding Ditto in the framework



Ditto **functionality customization**



Checking **results**

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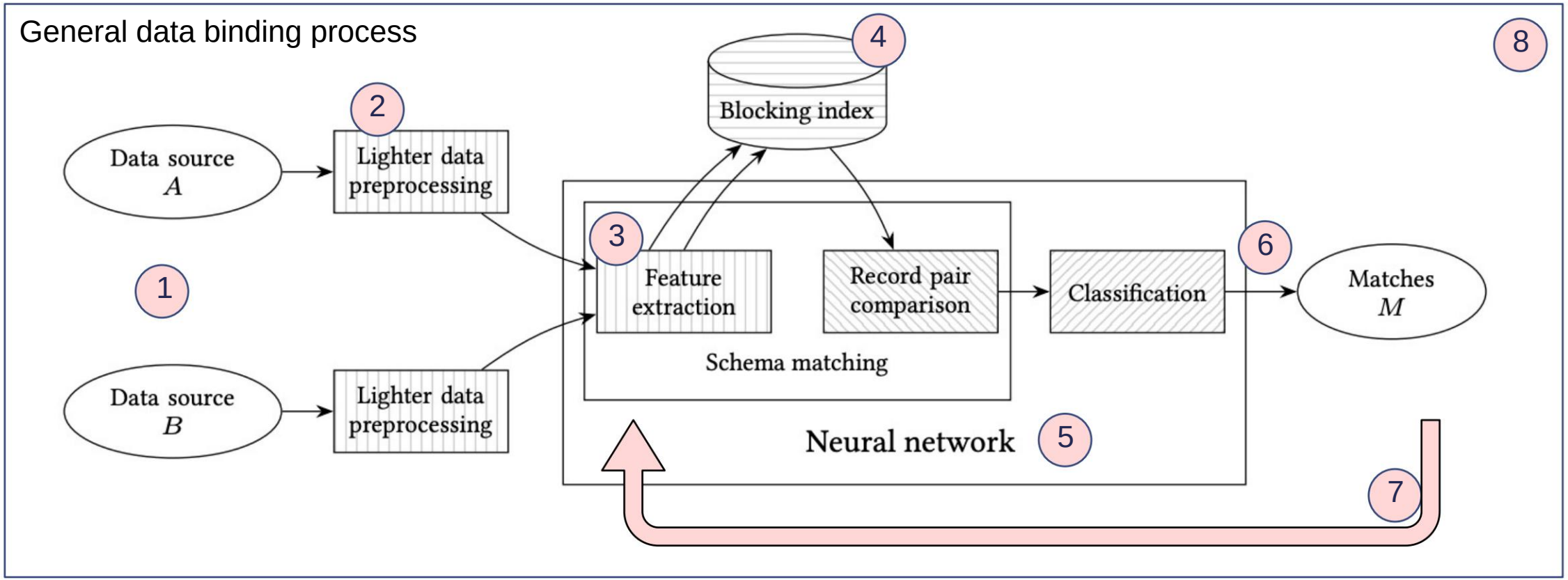
IMPLEMENTATION



1 Reading data from database

2 Add data preprocessing

3 Attributes selection

4 Pre-screening threshold manual entry

5 Subject area and model choice

6 Writing bindings to database

7 Iterative operation

8 Learning simulator and visualization

Checking record pairs

Home screen

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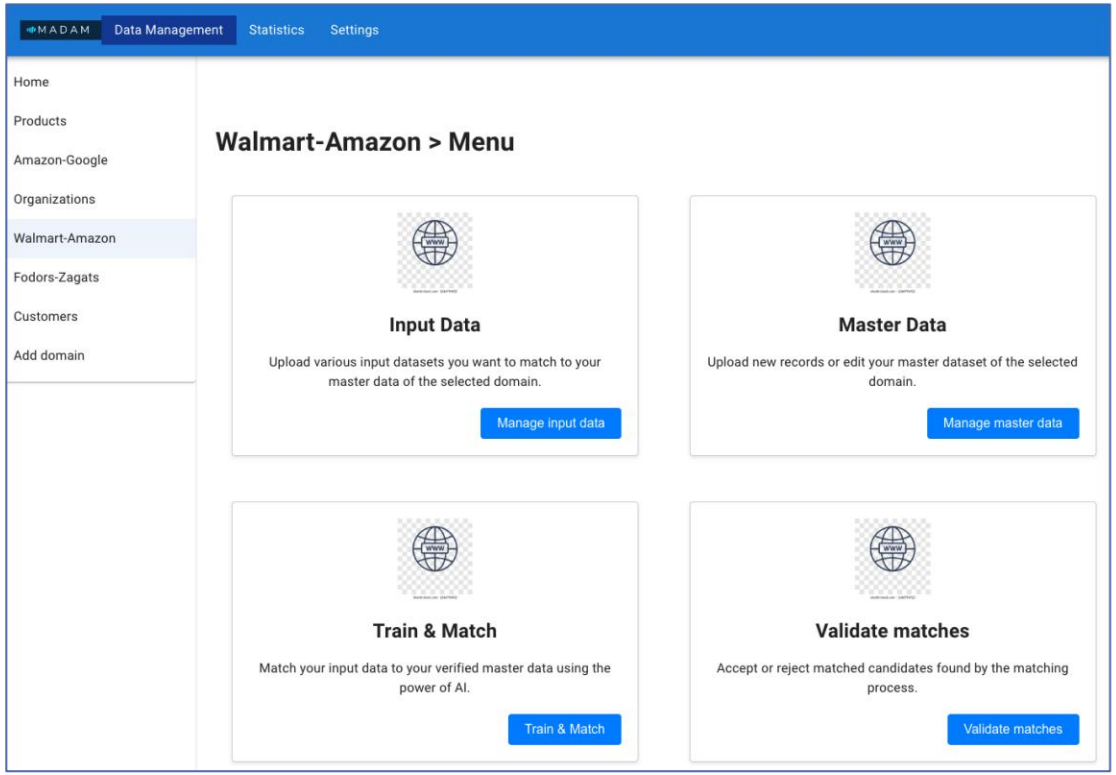
IMPLEMENTATION

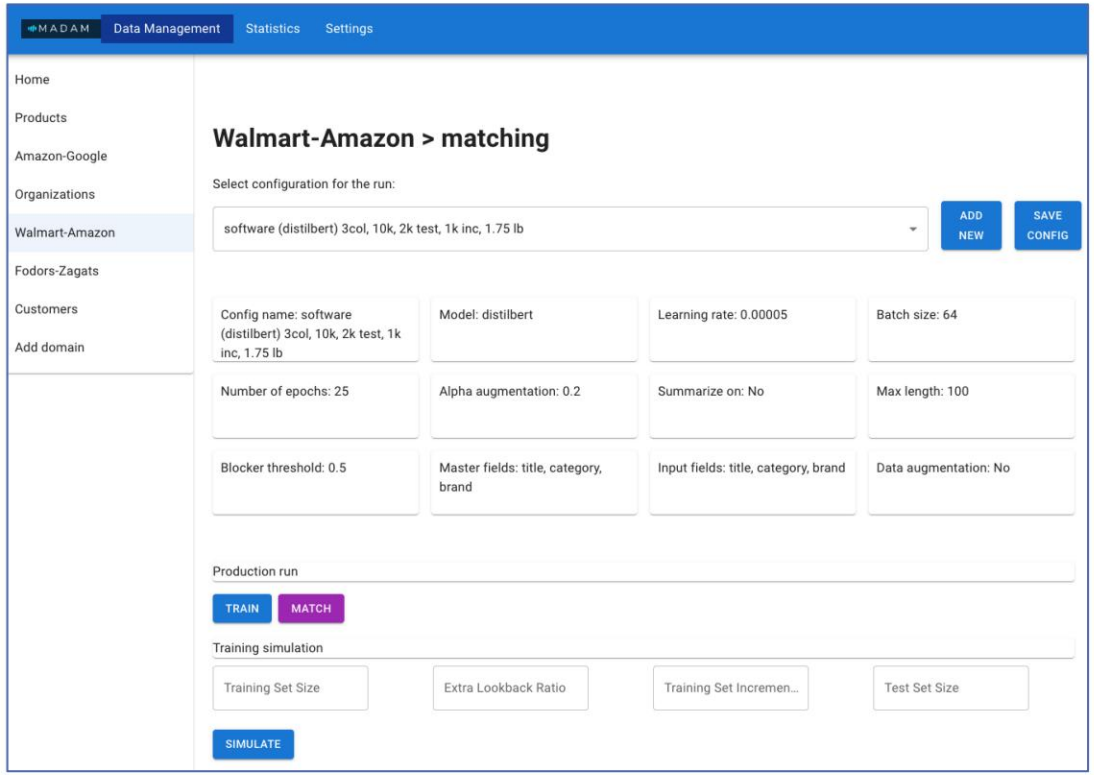
Topic area menu

- Features available for each domain:
  data management, record linking, verification



Data binding interface

- choosing learning and inference parameters
- teaching, record linking, simulation



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IMPLEMENTATION

Simulated teaching results

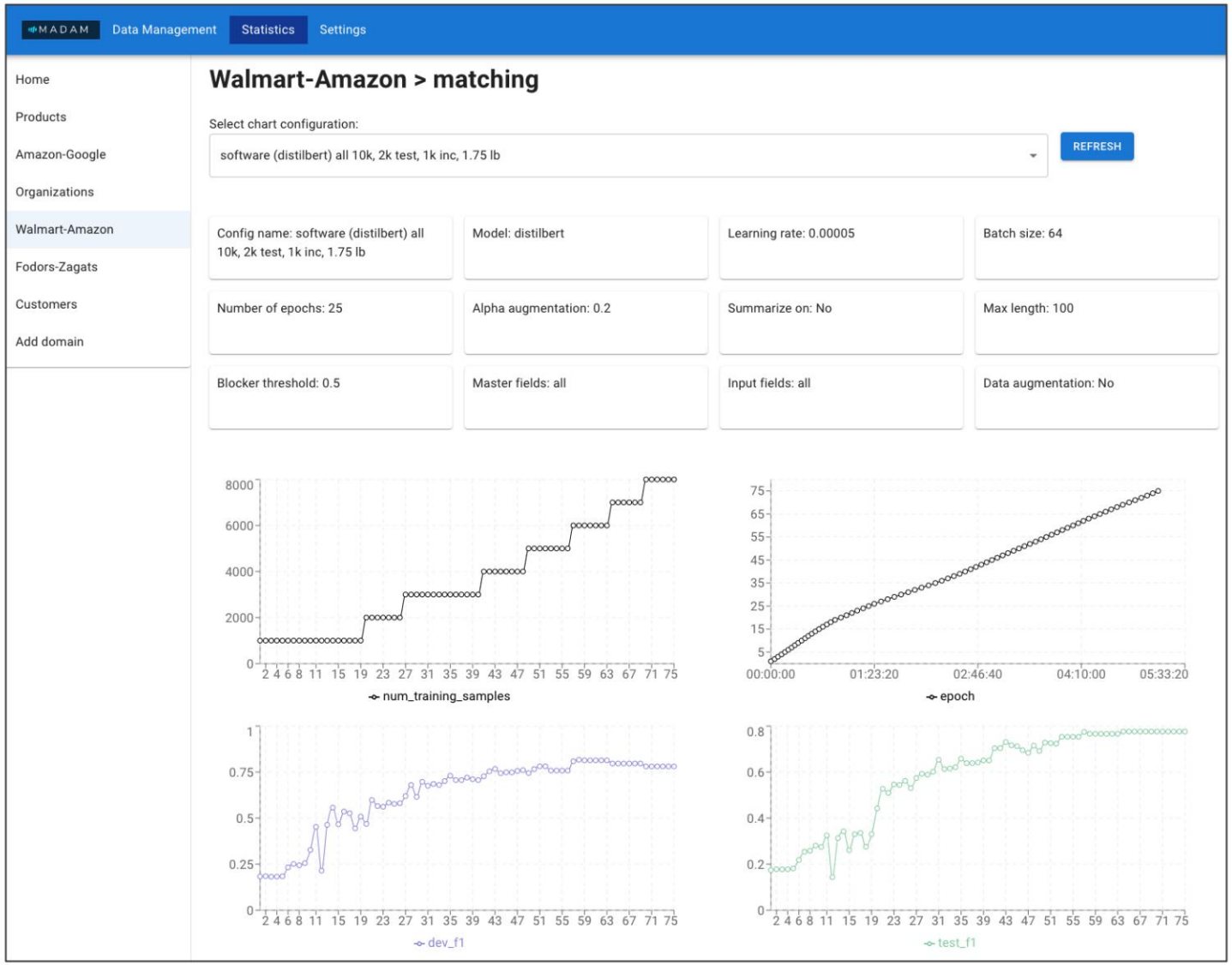
Visualizing the effectiveness of a model in a given domain with different settings.

Can be set:

- Data set size • Constant test set size
- Data set arriving with new iterations size •

Considered examples from the previous iteration

Available diagrams: • Training set size • Training process time • F1 score of the given training set • F1 score of the test set • (Precision) • (Recall)



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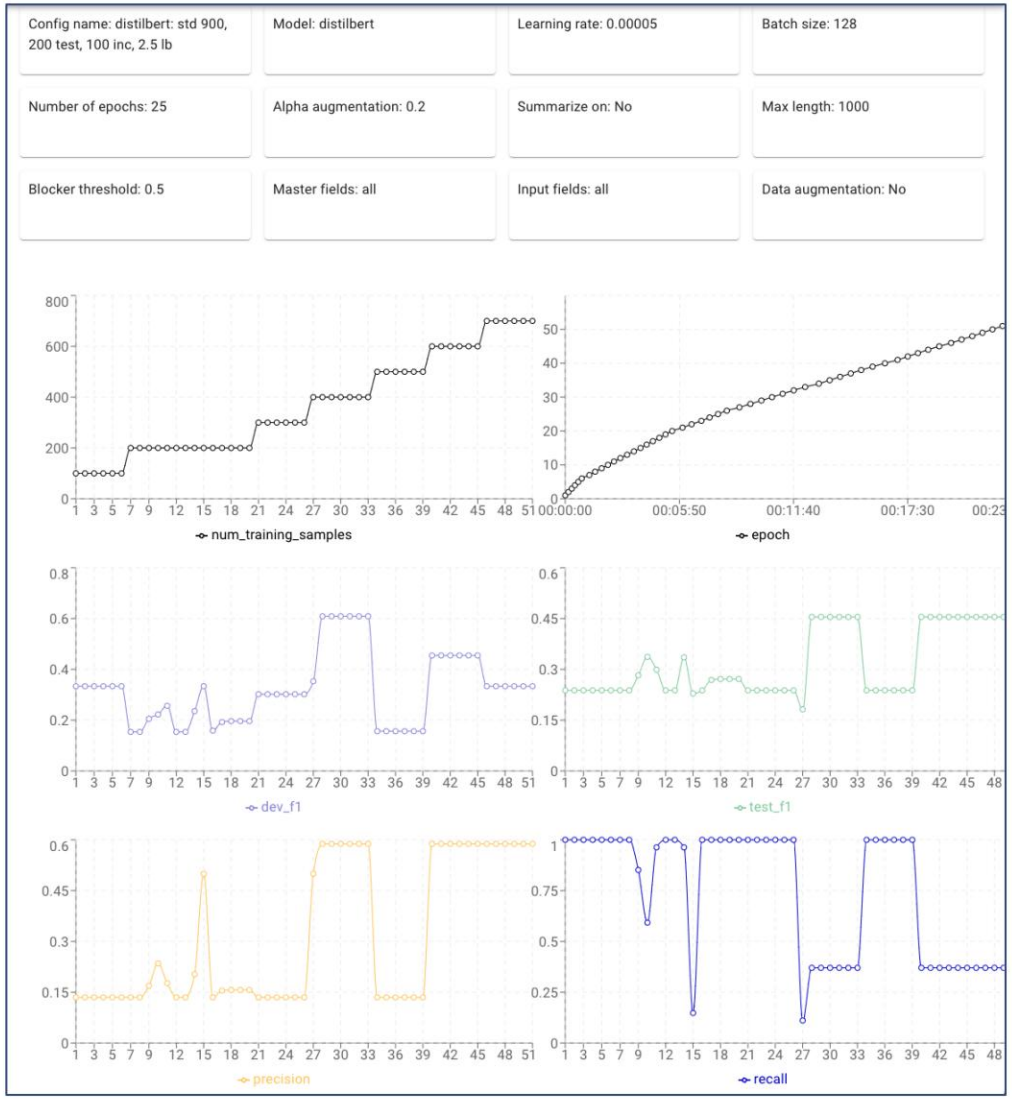
RESULTS

Amazon-Google benchmark dataset

Fodors-Zagats benchmark dataset



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THANK YOU FOR YOUR ATTENTION!

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APPENDIX

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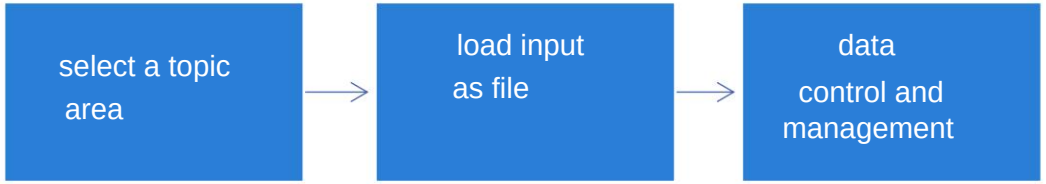
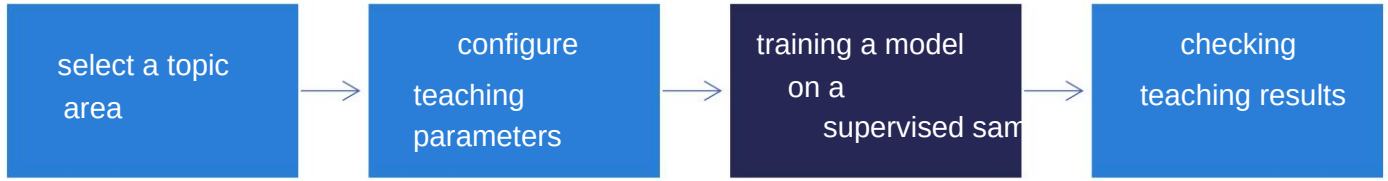
DESIGNED OPERATION

The data binding process works iteratively

1. A new data set is added to the external
   to be integrated into the data source

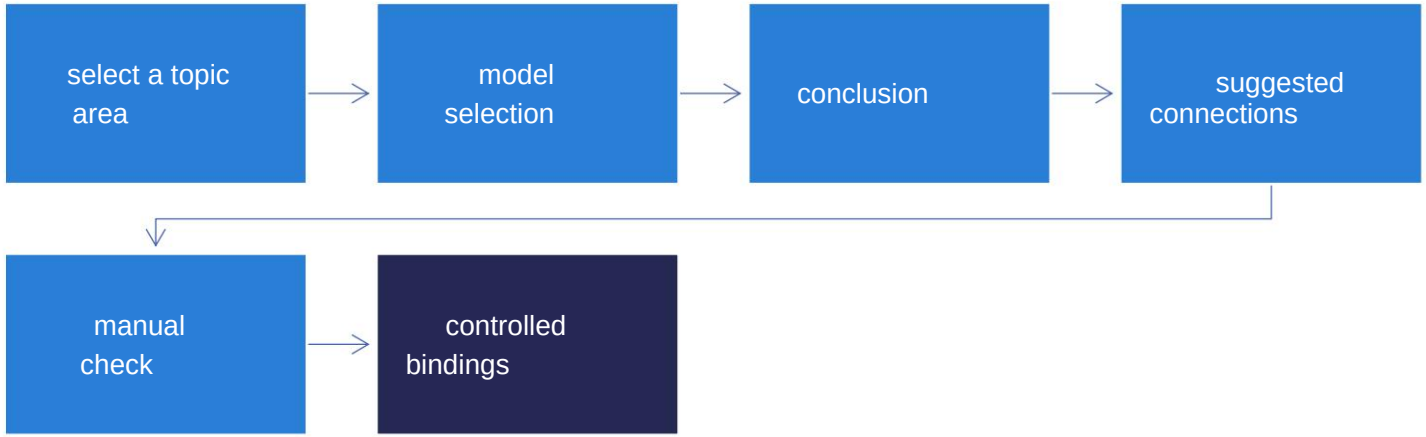


2. Fine-tuning the model based on the previous
   based on verified associations



3. Model for the topic area
   generates binding suggestions
   providing them with confidence values

4. User reviews and accepts or
   rejects the bindings



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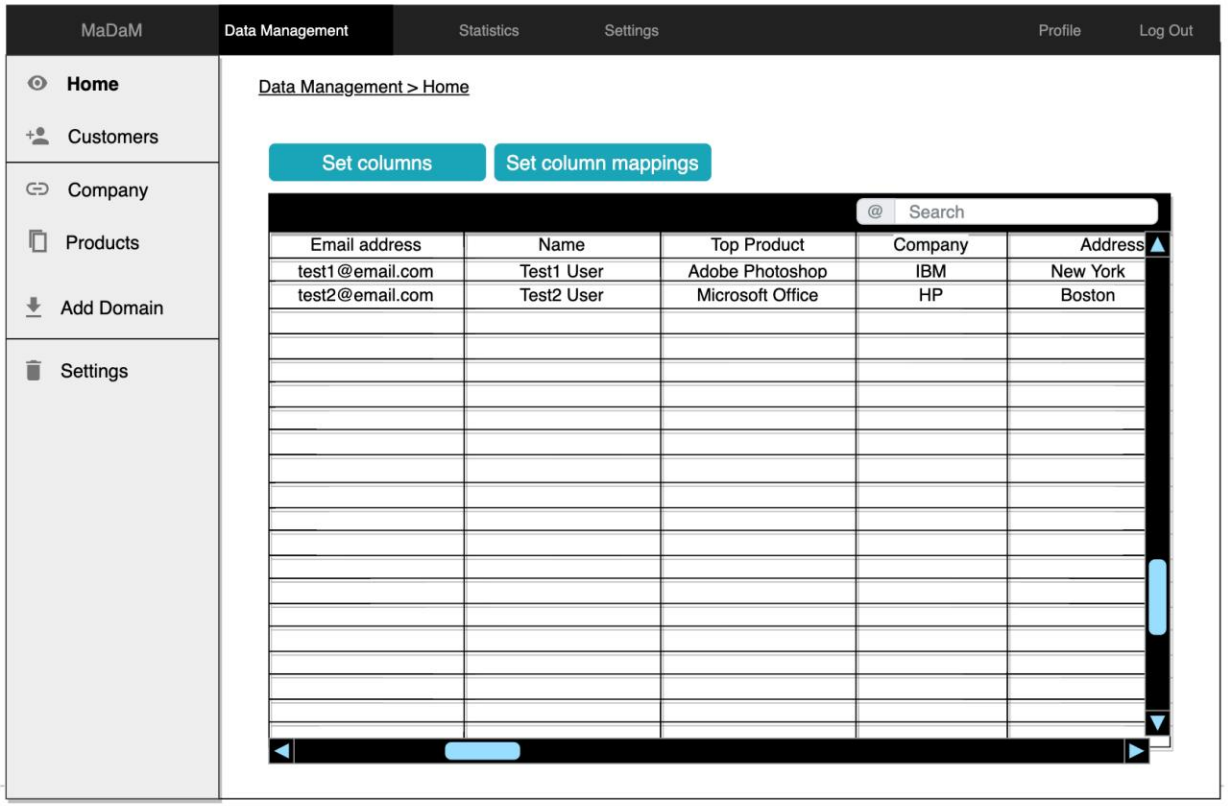
USER INTERFACE DESIGN

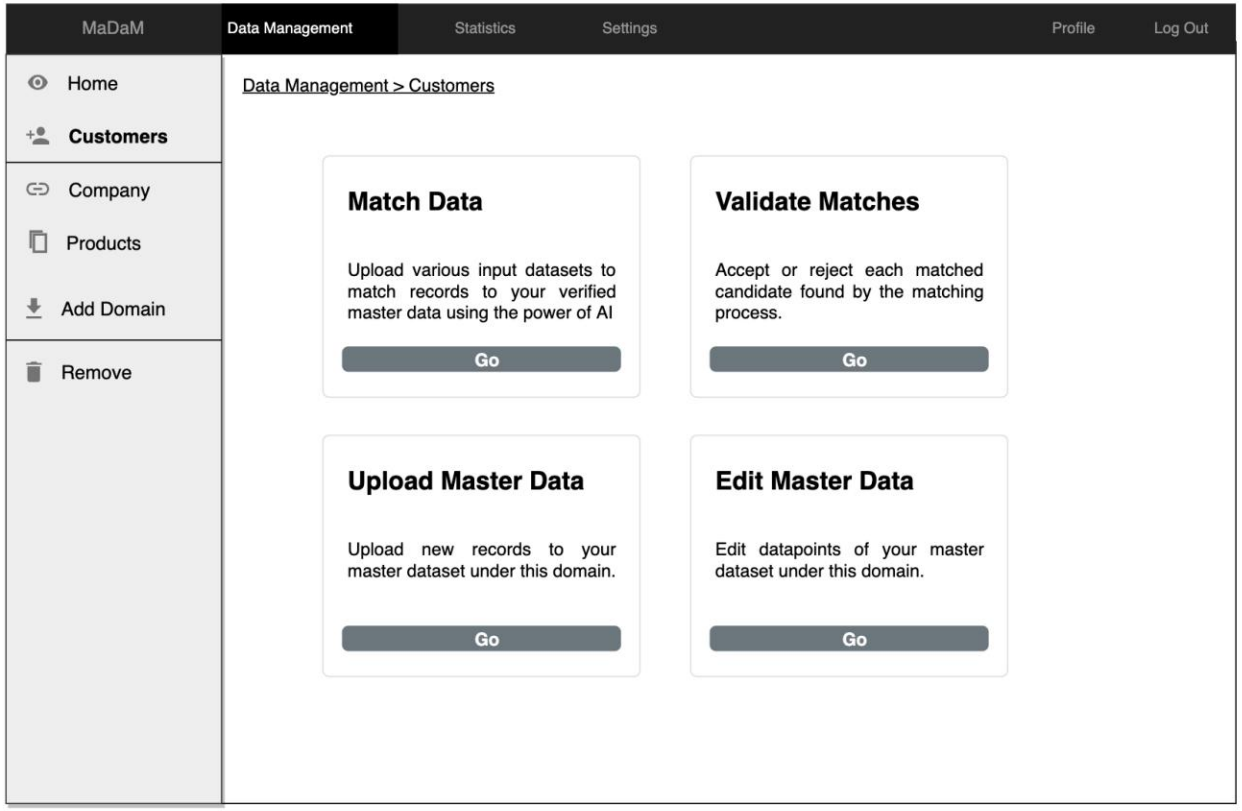
Home screen

• display of data points obtained by linking • display of standardized, standard values



Topic area menu

• to call functions available for each domain: record linking, verification, master data management



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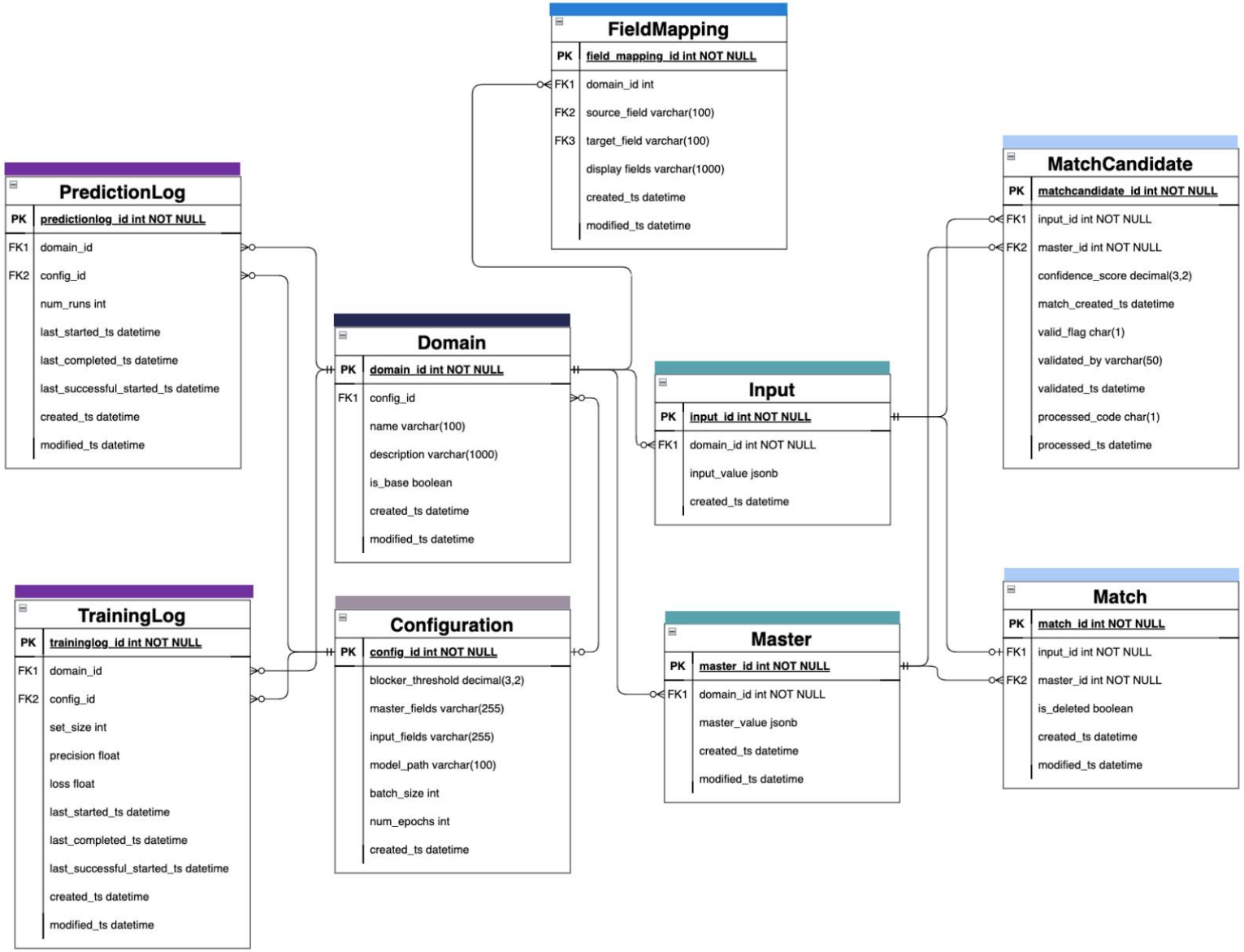
DATABASE PLAN

A database consisting of 9 tables, which can be grouped as follows:

- storage of inputs and master data
- data organization
  by subject areas
- preservation of bindings
- binding logic settings
- running logs
- data display settings



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TECHNOLOGY CHOICE

Python

- excellent applicability in the fields of data science and web development
- easy integration with DITTO with module

React

- modern front-end technology
- good performance
- easy to learn
- large user base

FastAPI

- modern web framework
- easy to use
- REST compatibility
- asynchronous operation

PostgreSQL

- relational database of record for easy querying of connections
- JSON type field for storing data with various structures